

STATSITICAL ANALYSIS OF ON-LINE LEARNING

OPTIMAL AND SEMI-OPTIMAL STOCHASTIC GRADIENT

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<https://noboru-murata.github.io/>

Introduction

- batch and on-line learning

Problem Formulation

- statistical properties of batch learning

- optimal learning rate for on-line learning

Illustrative Example

- Elo rating system

- restricted gradient problem

Conclusion

INTRODUCTION

notation:

- data: i.i.d.~ observations from ground truth distribution P

$$Z_1, Z_2, \dots, Z_t, \dots \sim^{\text{i.i.d.}} P$$

- learning machine: specified by a finite dimensional parameter

$$\theta \in \Theta \subset \mathbb{R}^m$$

- loss function: penalty of machine θ for a given datum z

$$l(z; \theta) \quad (\text{a smooth function with respect to } \theta)$$

for example

$$l(z; \theta) = -\log p(z; \theta) \quad \text{negative log loss}$$

$$l(z; \theta) = |y - f(x; \theta)|^2 \quad \text{squared loss for } z = (x, y) \text{ (location model)}$$

- batch learning: minimize the empirical loss

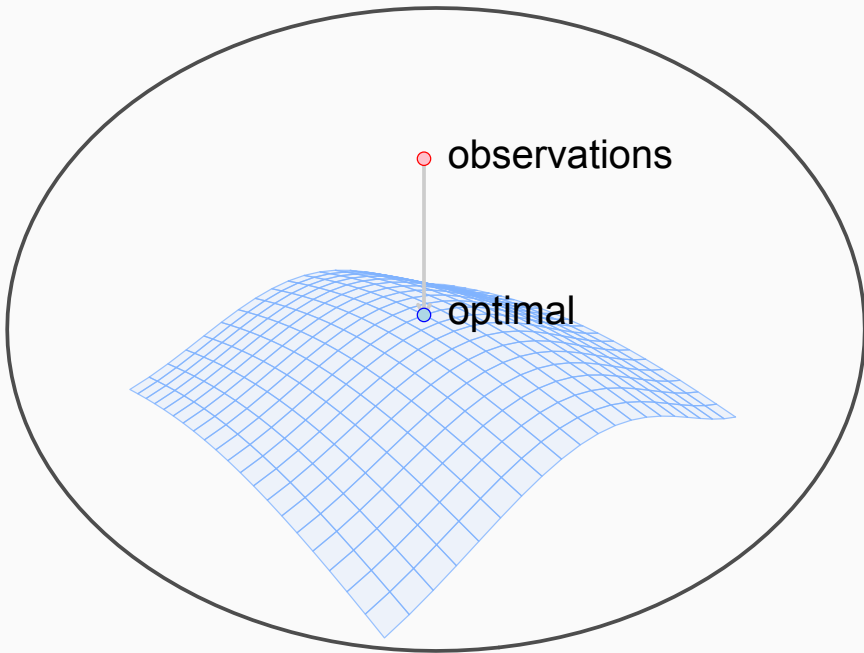
$$\hat{\theta}_t = \arg \min_{\theta} \hat{L}_t(\theta),$$

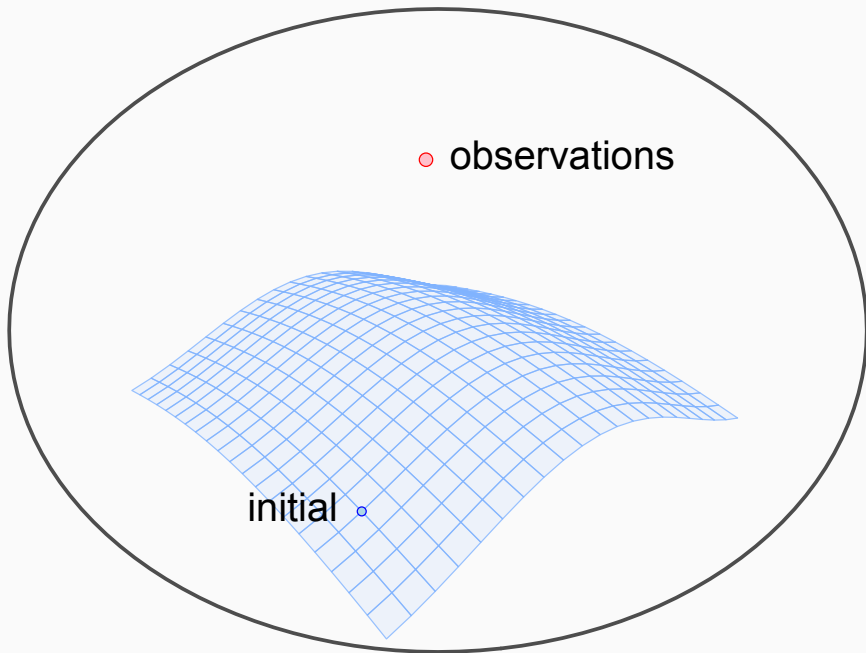
- on-line learning: update sequentially with a datum sampled at each time (or resampled from pooled data)

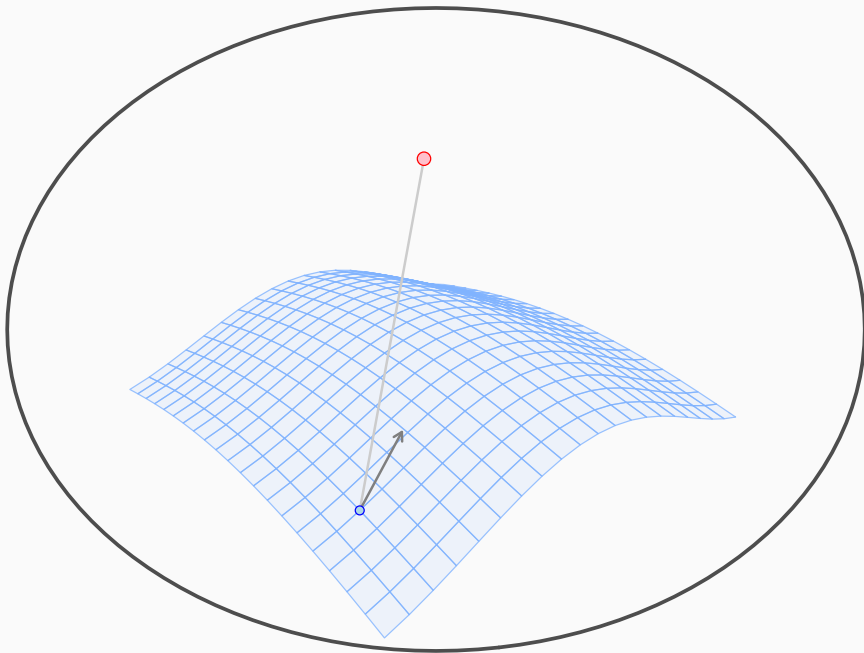
$$\theta_t = \theta_{t-1} - \Phi_t \nabla l(z_t; \theta_{t-1}),$$

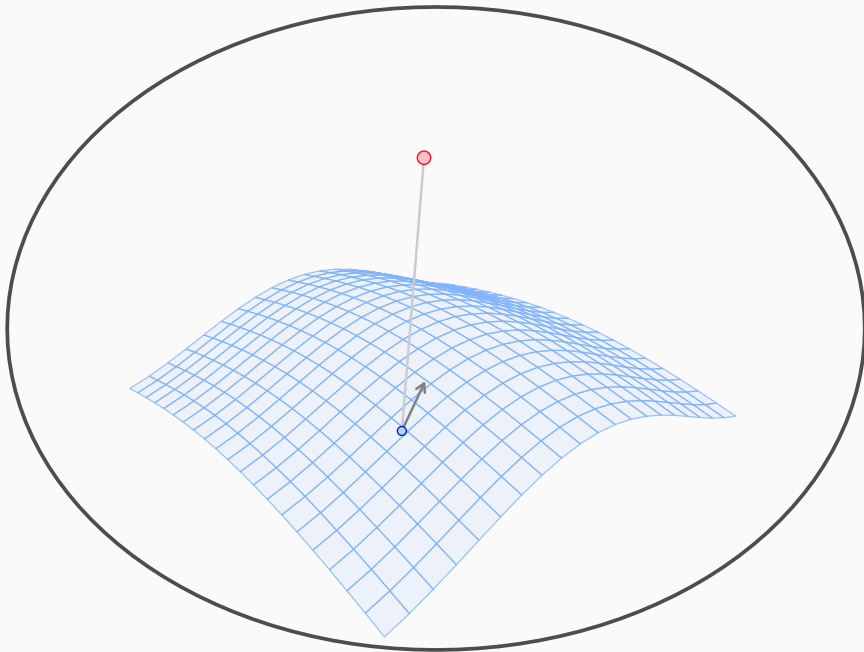
where ∇ denotes the gradient with respect to θ , and Φ is a matrix which controls the rate of convergence.

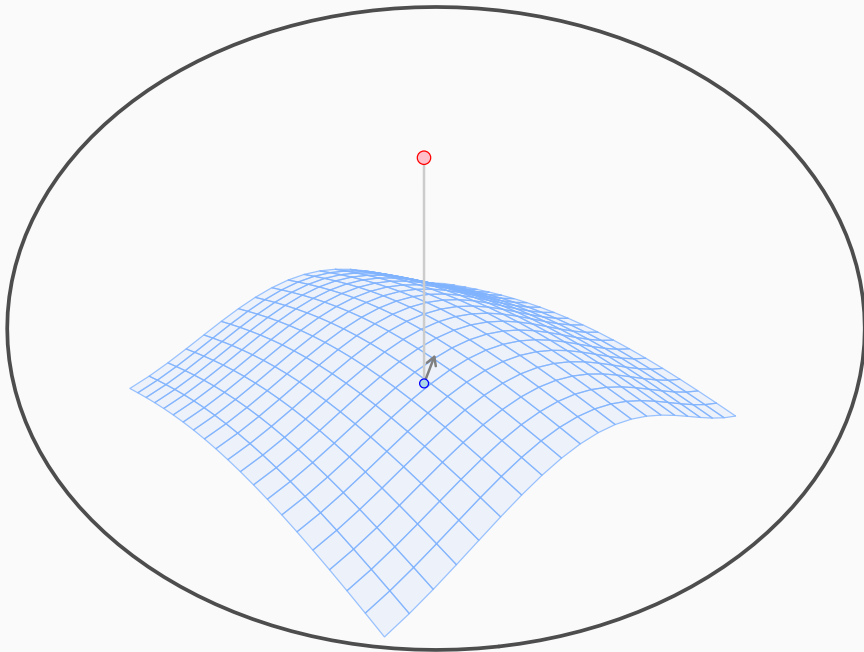
- batch learning:
 - pros** can adopt wide class of loss functions
 - cons** shows slow convergence
 - may have many local minima
 - should store all the observations
- on-line learning:
 - pros** do not have to store all the observations
 - (good for massive data stream)
 - can escape from local minima
 - can follow the change of true distributions
 - cons** should control learning rate ε properly
 - (do not converge with constant ε)

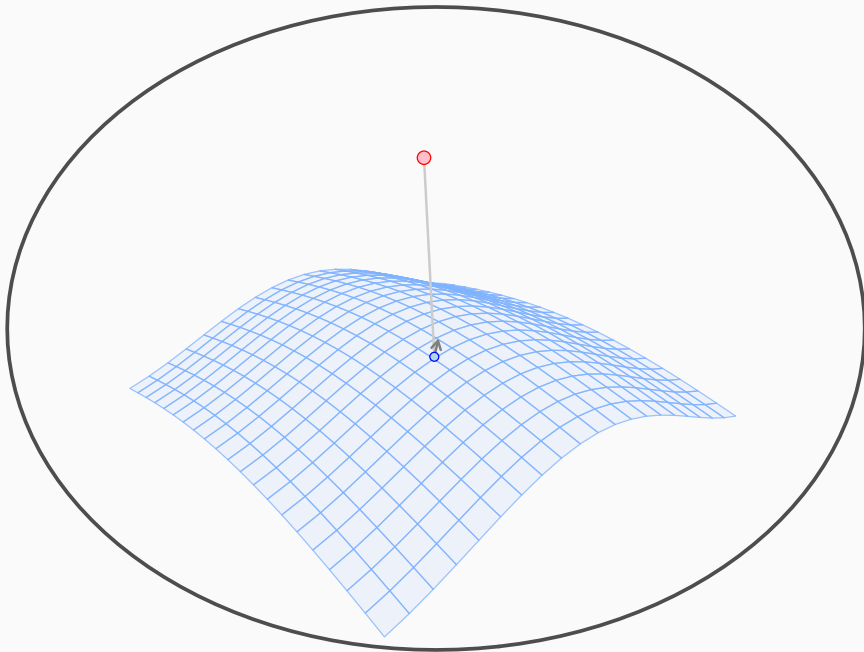


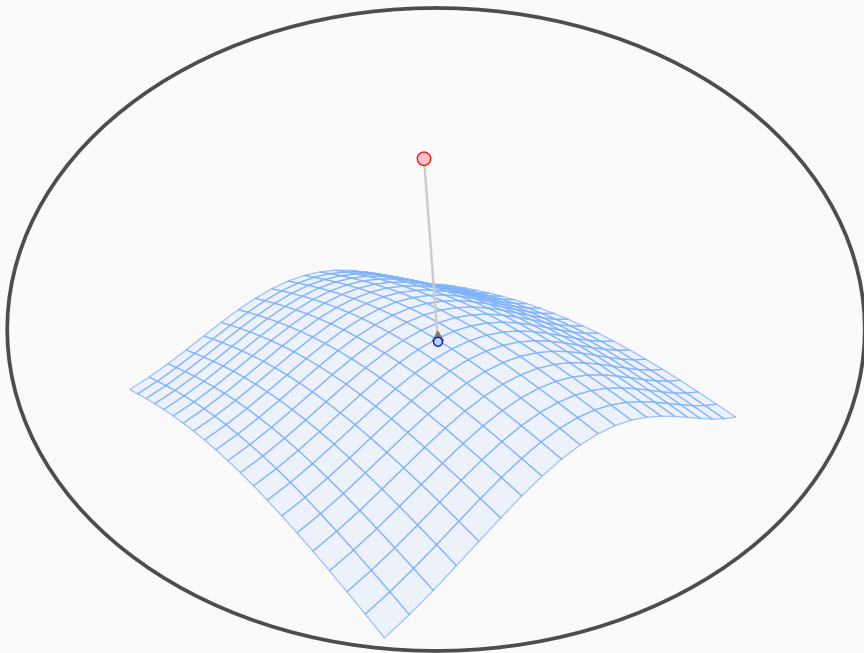


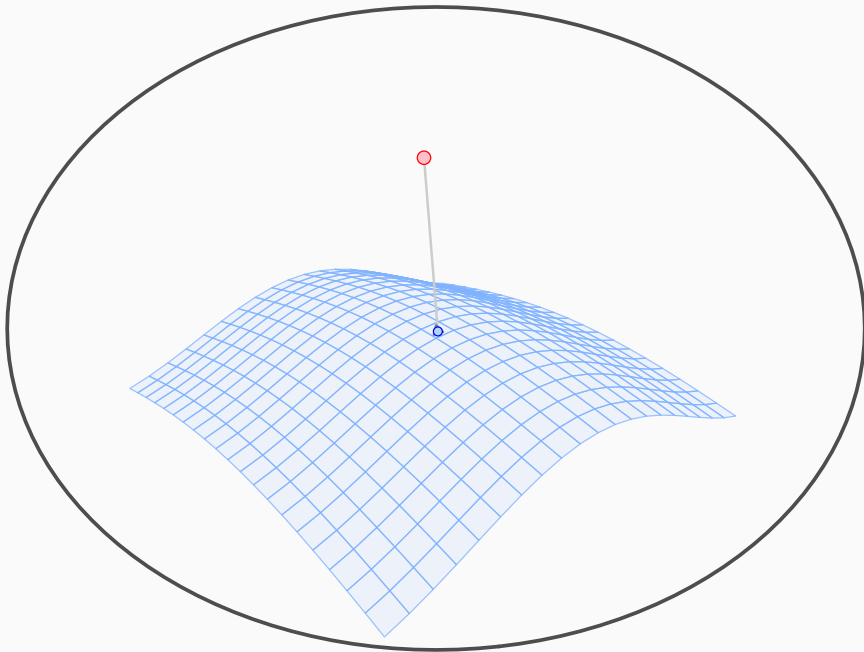


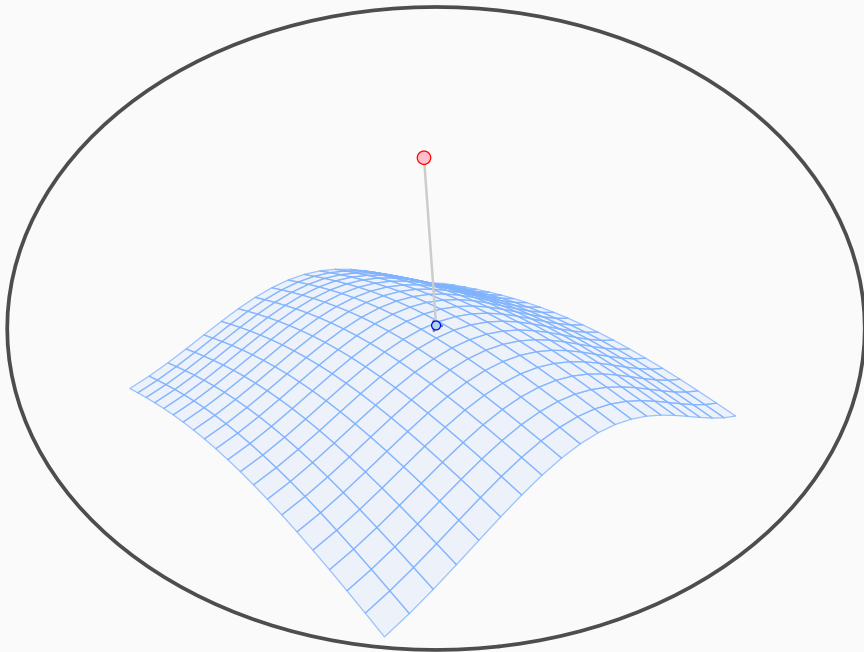


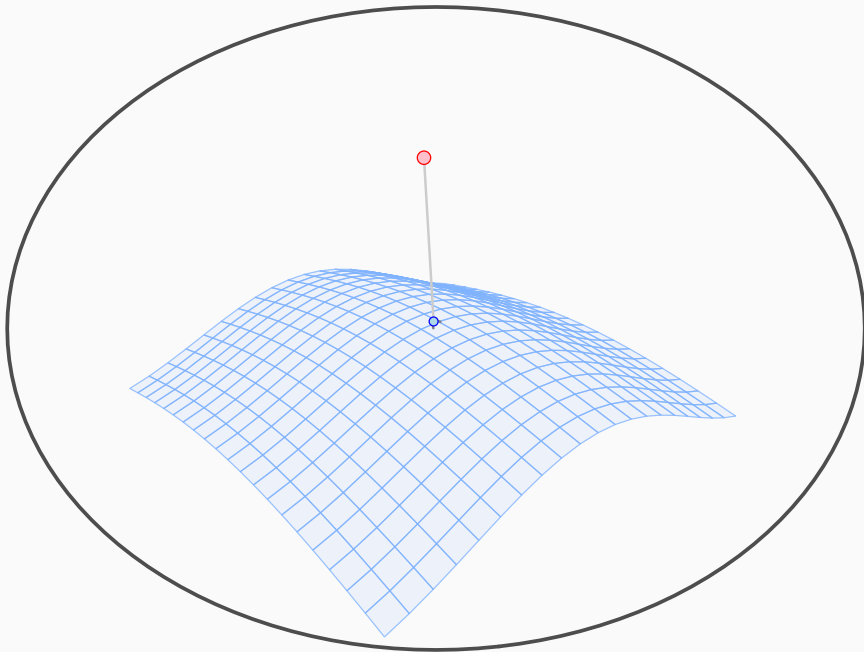




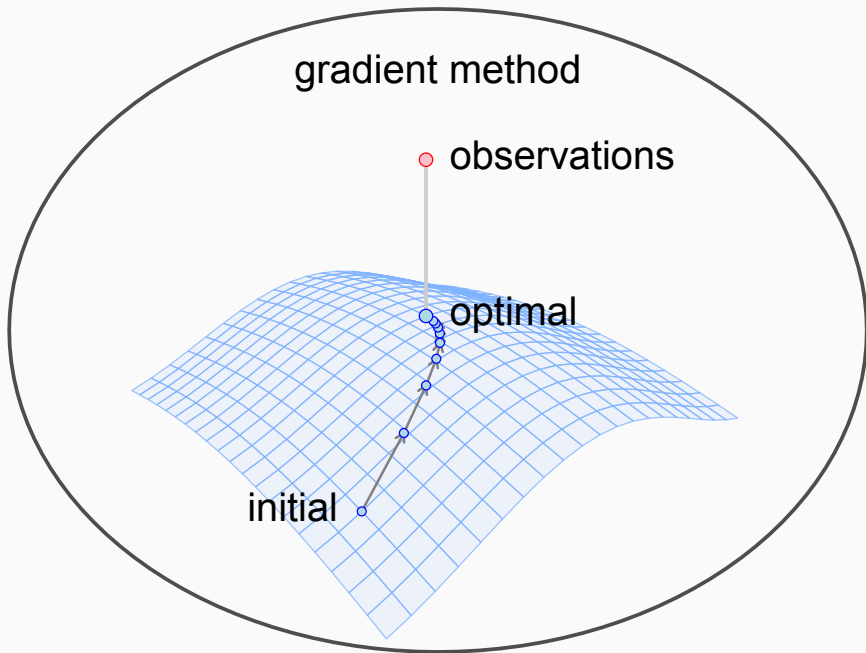


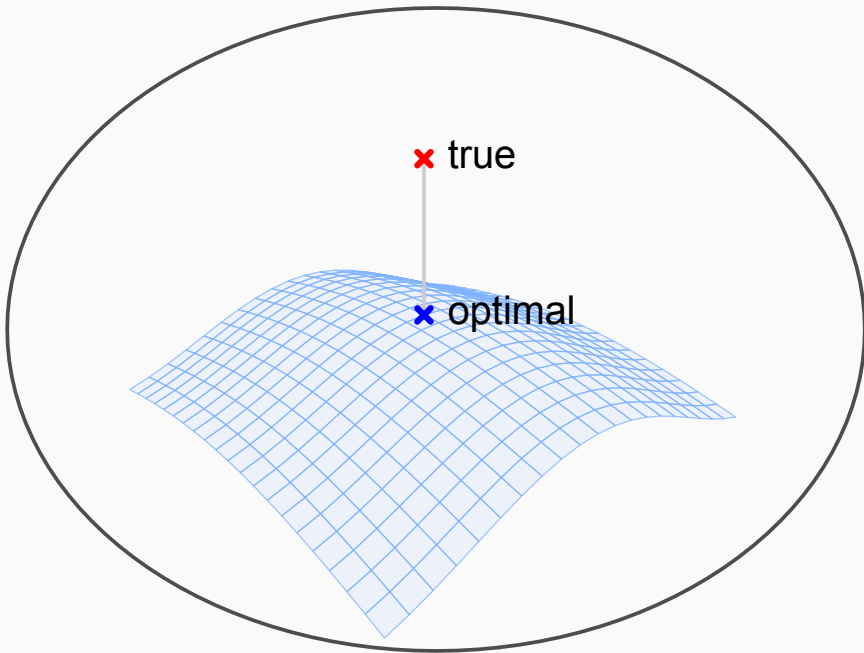


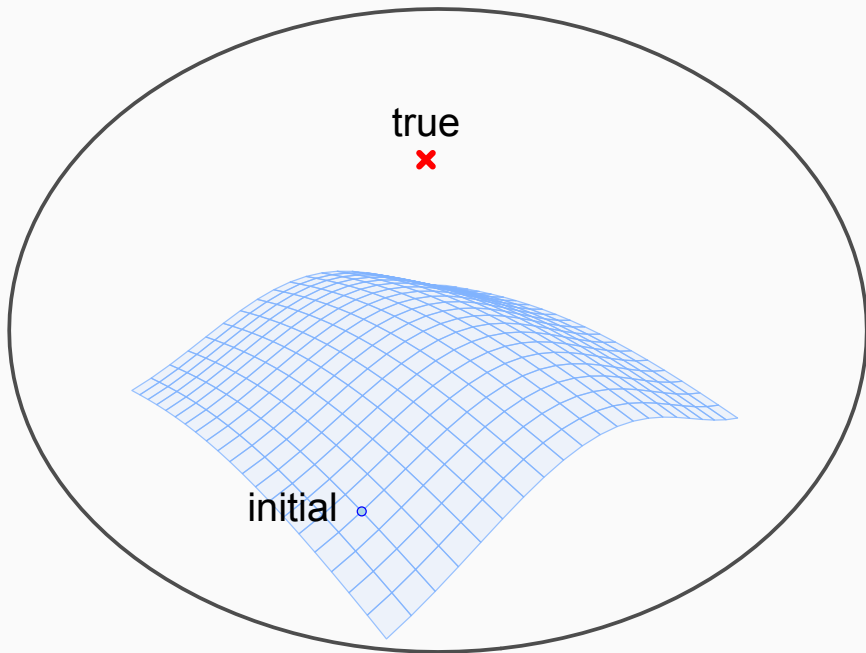


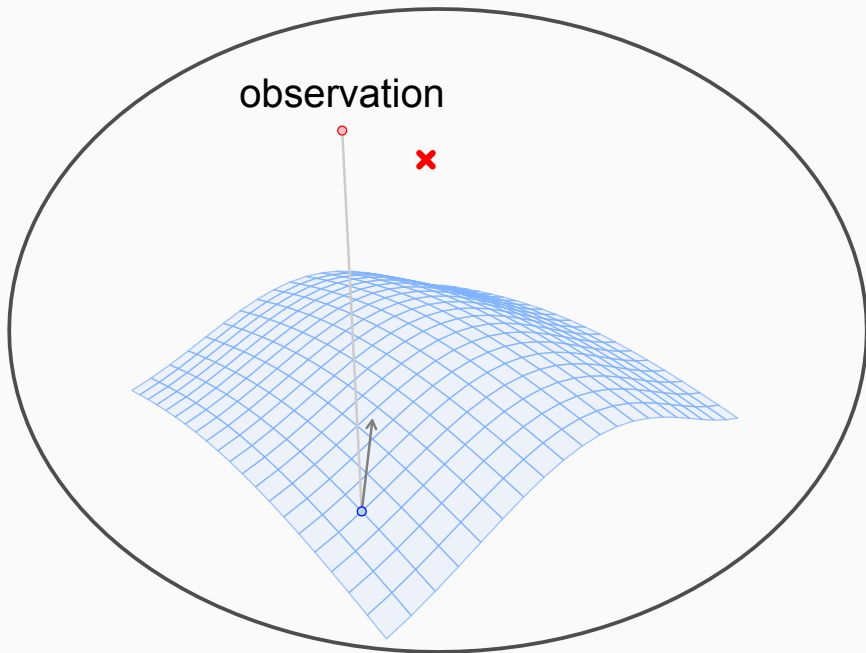


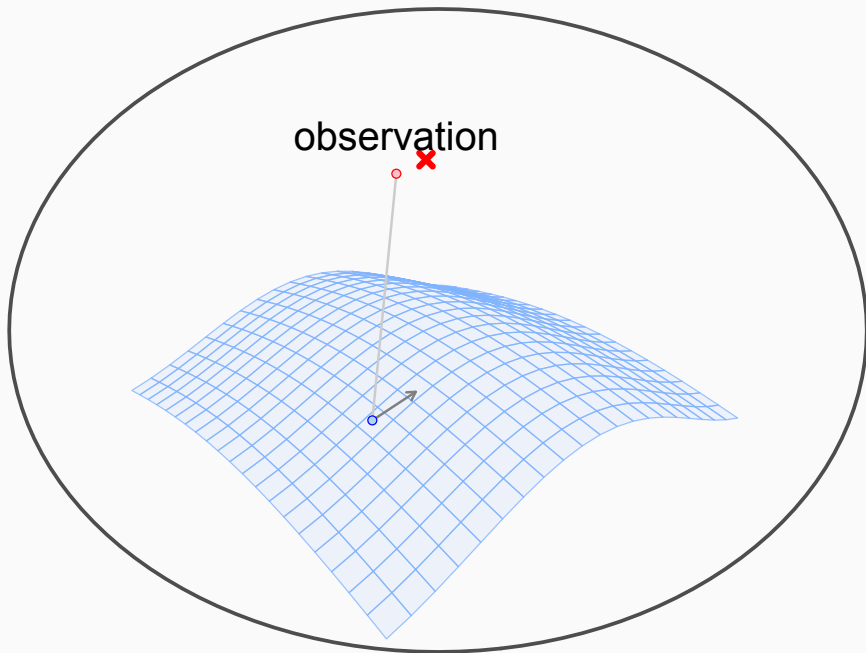
gradient method

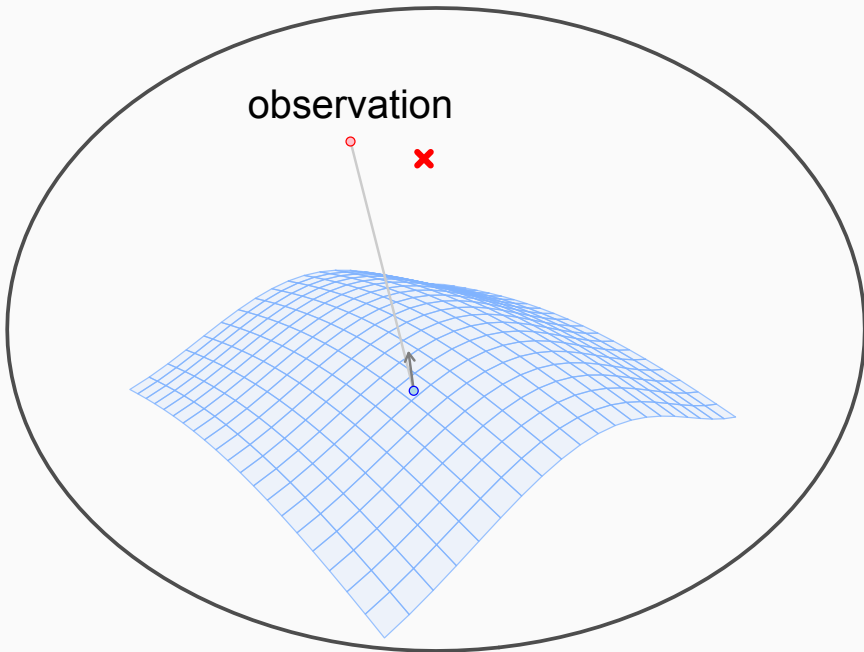


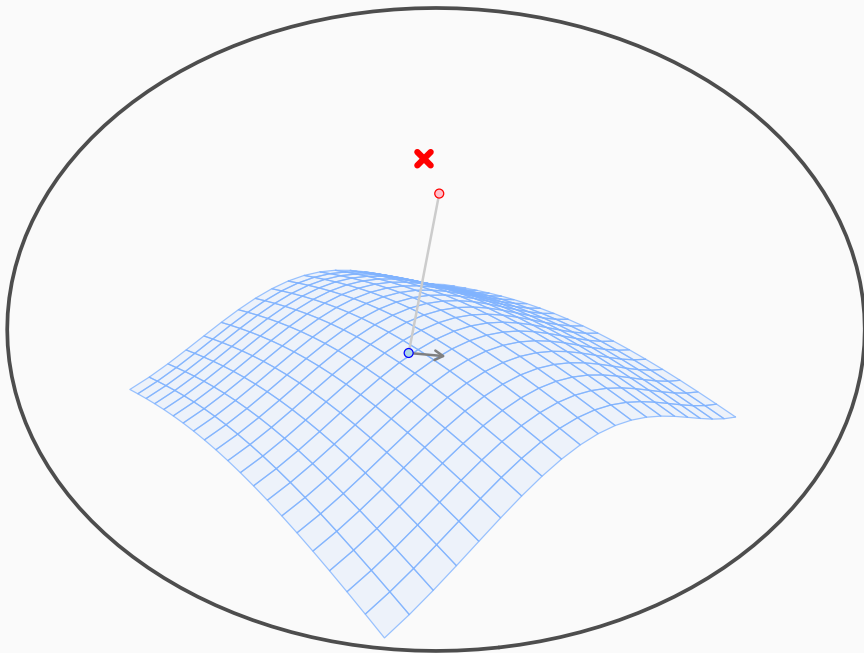


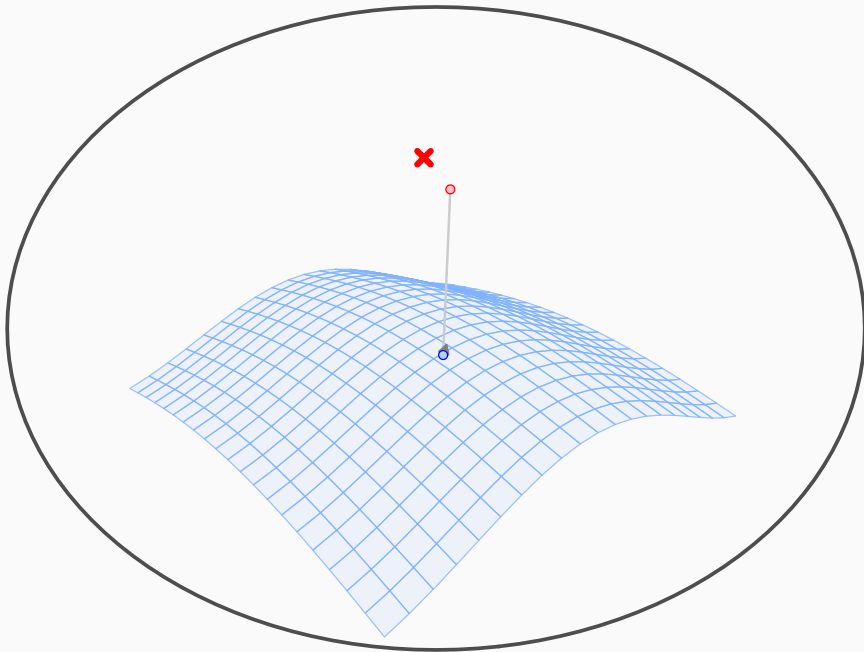


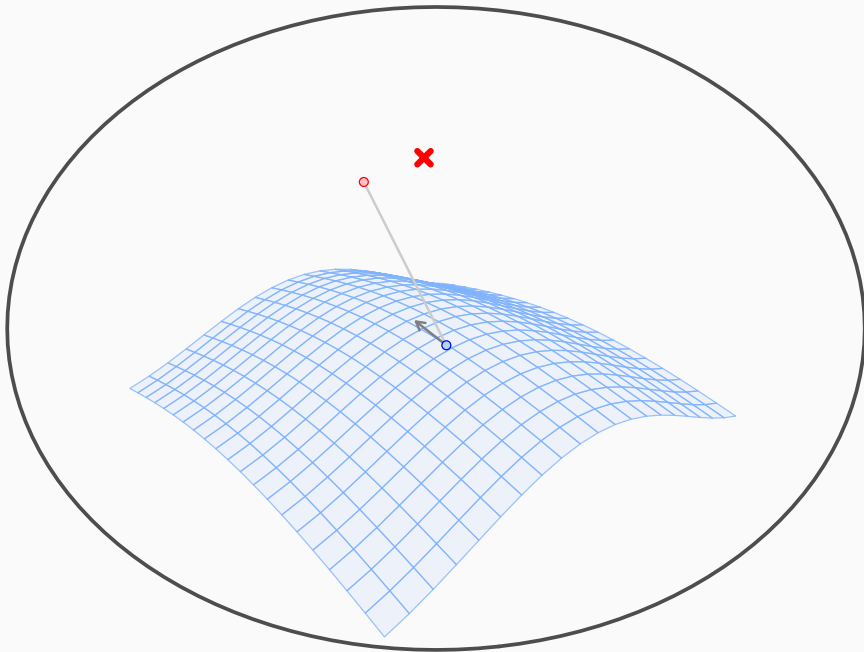


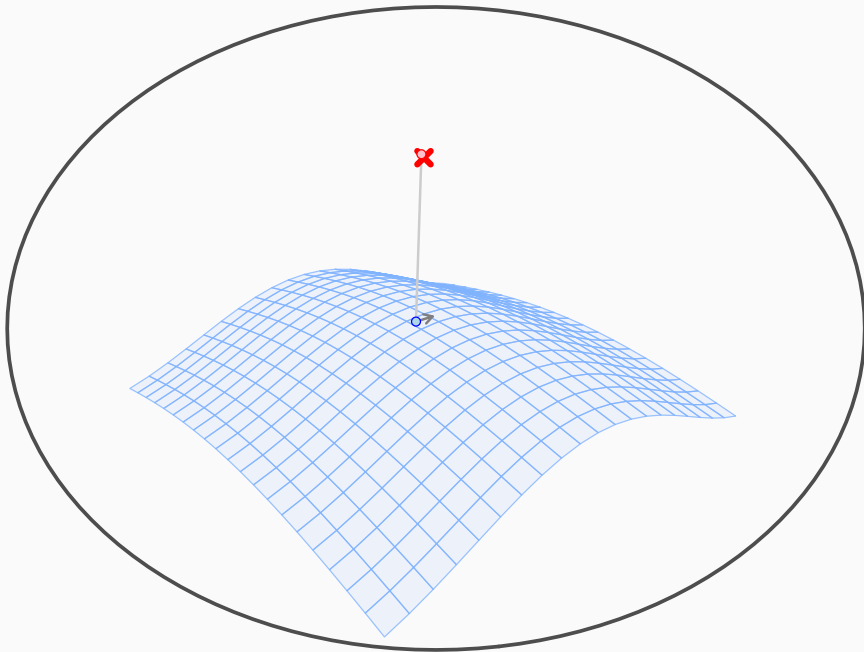


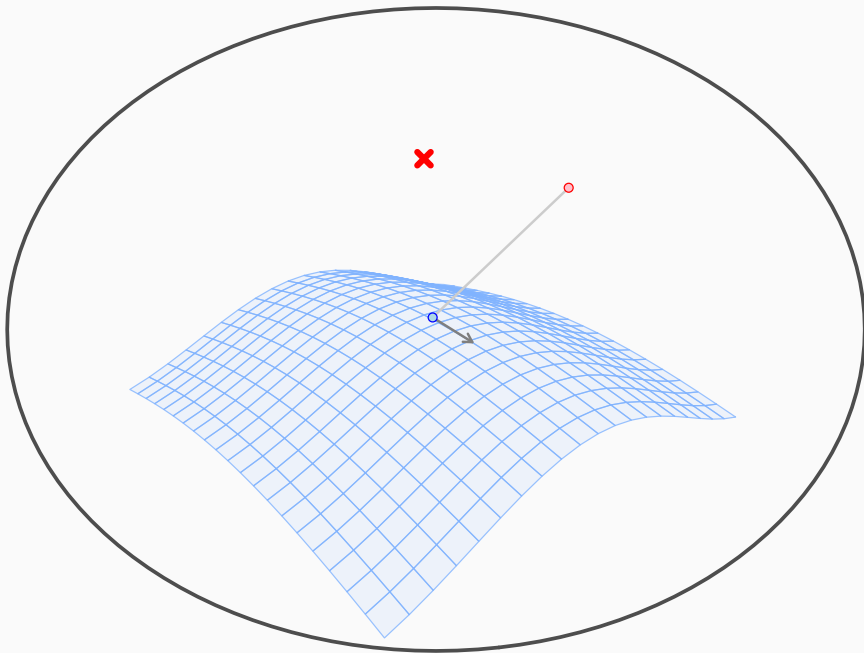




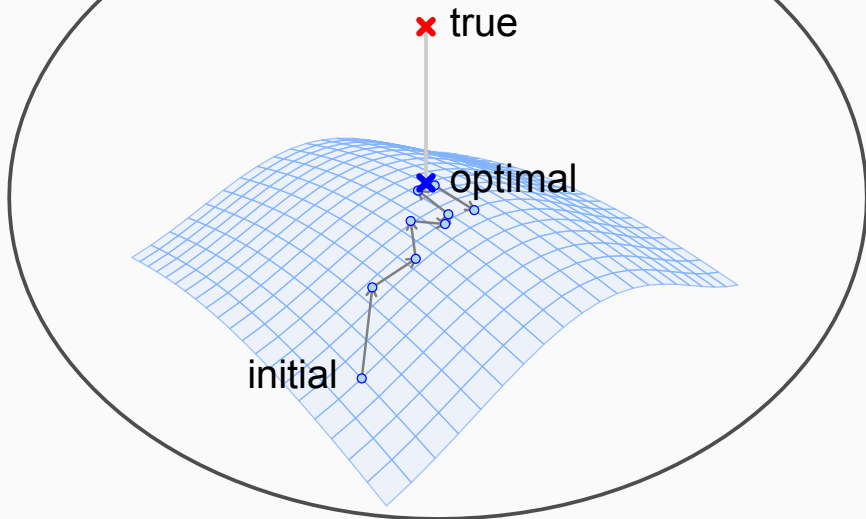








stochastic approximation



- is on-line learning inferior to batch?
- how on-line estimators behave?
- what are good learning parameters?

PROBLEM FORMULATION

The expectation of the population loss is asymptotically given by

$$\mathbb{E}\left[L(\hat{\theta}_t)\right] = L(\theta_*) + \frac{1}{2t} \text{tr} GH^{-1} + o\left(\frac{1}{t}\right),$$

where the expectation is taken with respect to D_t .

The variance is asymptotically given by

$$\mathbb{V}\left[L(\hat{\theta}_t)\right] = \frac{1}{2t^2} \text{tr } GH^{-1}GH^{-1} + o\left(\frac{1}{t^2}\right).$$

Theorem

The expectation of the empirical loss is asymptotically given by

$$\mathbb{E}[\hat{L}_t(\hat{\theta}_t)] = L(\theta_*) - \frac{1}{2t} \text{tr} GH^{-1} + o\left(\frac{1}{t}\right),$$

where the expectation is taken with respect to D_t .

The variance is asymptotically given by

$$\mathbb{V}[\hat{L}_t(\hat{\theta}_t)] = \frac{1}{t} \mathbb{V}_{Z \sim P} [l(Z; \theta_*)] + o\left(\frac{1}{t}\right).$$

- generalization error:

$$\mathbb{E}\left[L(\hat{\theta}_t)\right] = L(\theta_*) + \frac{1}{2t} \text{tr } GH^{-1} + o\left(\frac{1}{t}\right),$$

- training error:

$$\mathbb{E}\left[\hat{L}_t(\hat{\theta}_t)\right] = L(\theta_*) - \frac{1}{2t} \text{tr } GH^{-1} + o\left(\frac{1}{t}\right),$$

Corollary (Akaike, 1974)

The generalization error is estimated from the training error by correcting the bias as

$$L(\hat{\theta}_t) = \hat{L}_t(\hat{\theta}_t) + \frac{1}{t} \text{tr } GH^{-1}.$$

In the case of the maximum likelihood estimation, if the ground truth is realized by θ_ ,*

$$L(\hat{\theta}_t) = \hat{L}_t(\hat{\theta}_t) + \frac{m}{t} \quad (m : \text{dim. of } \theta),$$

because $H = G$.

Lemma (Akahira & Takeuchi, 1981; Bottou & Le Cun, 2005)

Let $\hat{\theta}_{t-1}$ and $\hat{\theta}_t$ be estimates for D_{t-1} and $D_t = D_{t-1} \cup \{z_t\}$. Then

$$\hat{\theta}_t = \hat{\theta}_{t-1} - \frac{1}{t} \hat{H}_t^{-1} \nabla l(z_t; \hat{\theta}_{t-1}) + \mathcal{O}_p\left(\frac{1}{t^2}\right)$$

holds under some mild condition, where $\hat{H}_t$ is the empirical Hessian defined by

$$\hat{H}_t = \frac{1}{t} \sum_{z_i \in D_t} \nabla \nabla l(z_i; \hat{\theta}_{t-1}).$$

- batch learning:

$$\hat{\theta}_t = \hat{\theta}_{t-1} - \frac{1}{t} \hat{H}_t^{-1} \nabla l(z_t; \hat{\theta}_{t-1}) + (\text{higher order term})$$

- optimal on-line learning:

$$\theta_t = \theta_{t-1} - \frac{1}{t} \tilde{H}_t^{-1} \nabla l(z_t; \theta_{t-1}) + (\text{higher order term})$$

- optimal design: Newton-Raphson + $1/t$ -annealing

$$\Phi_t = \frac{1}{t} \hat{H}_t^{-1},$$

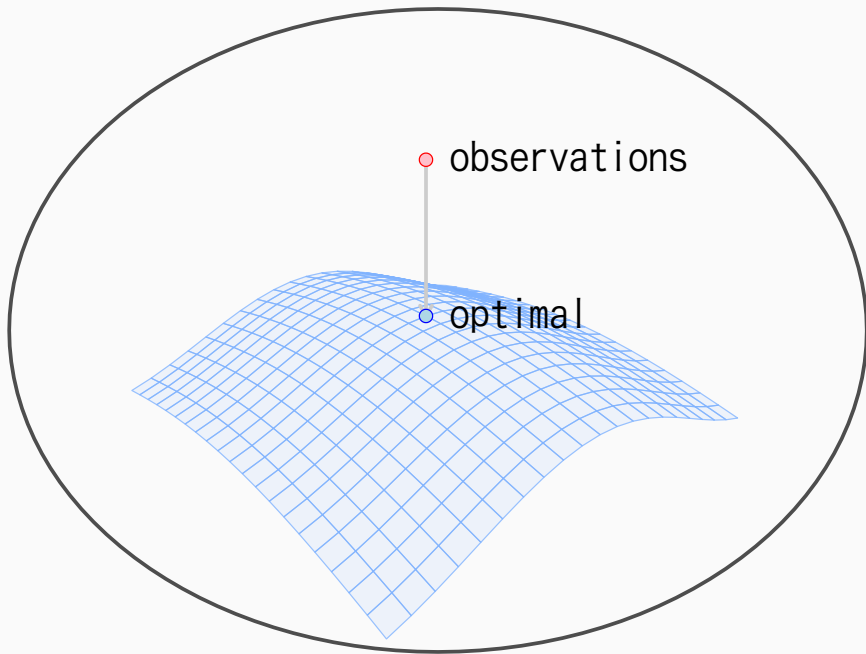
- on-line estimate of Hessian:
(MLE case; Bottou, 1998)

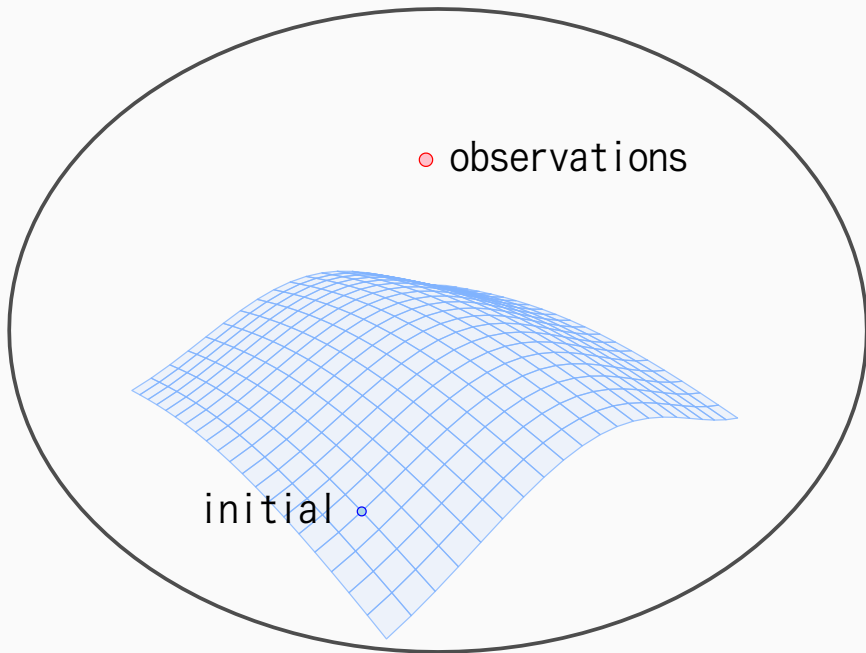
$$\Phi_t = \Phi_{t-1} - \frac{\Phi_{t-1} \nabla l \nabla l^\top \Phi_{t-1}}{1 + \nabla l^\top \Phi_{t-1} \nabla l}$$

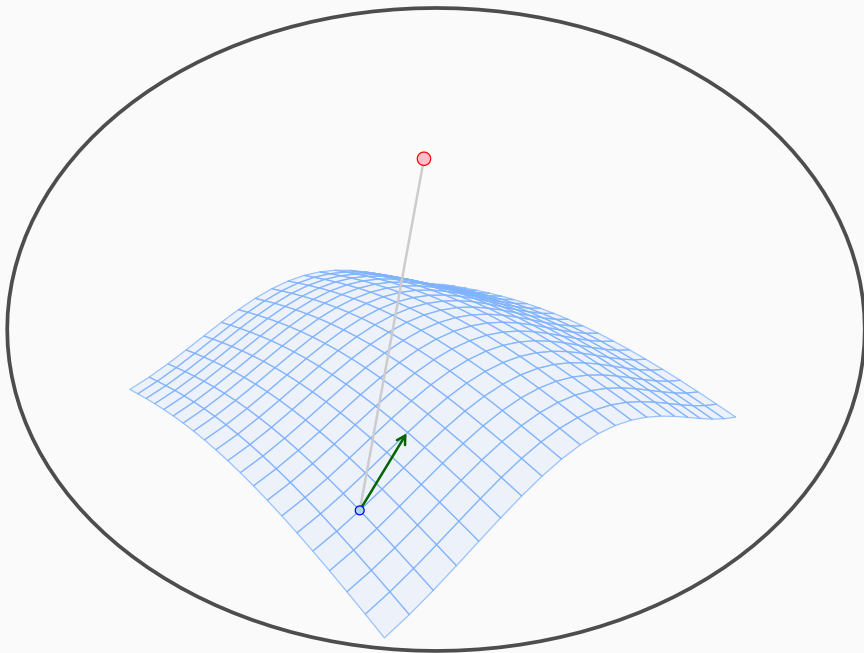
$$\text{where } \nabla l = \nabla l(z_t; \theta_{t-1})$$

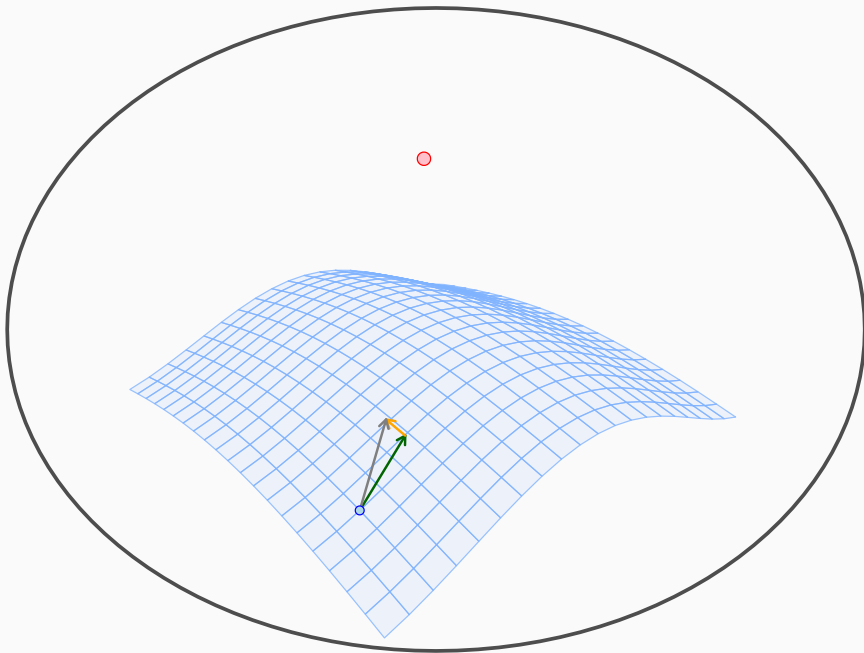
stochastic-BFGS (Nocedal et al, 2014), etc.

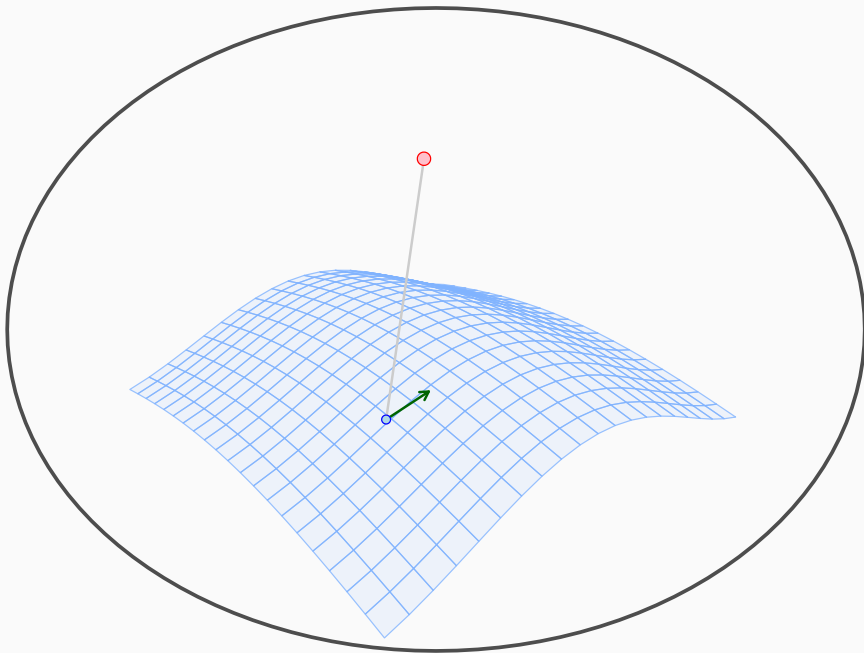
- rate of convergence: **equivalent with batch learning**
(NM, 1998; NM & Amari, 1999; Bottou & Le Cun, 2005)

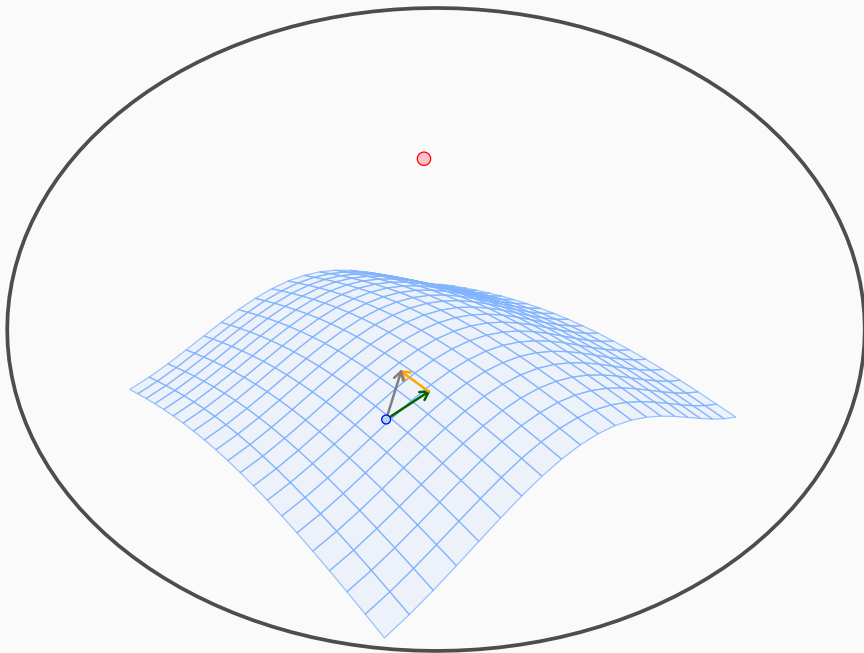


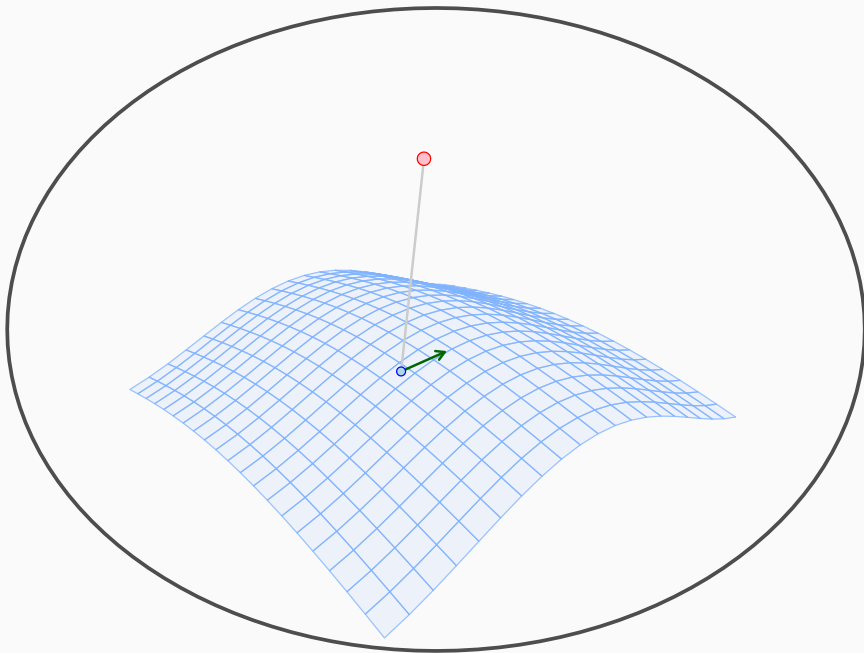


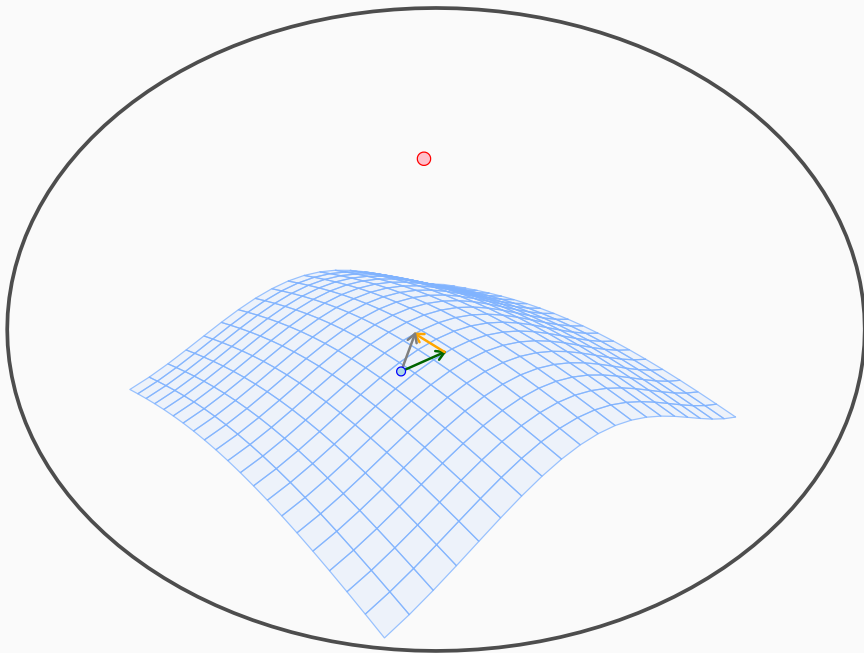


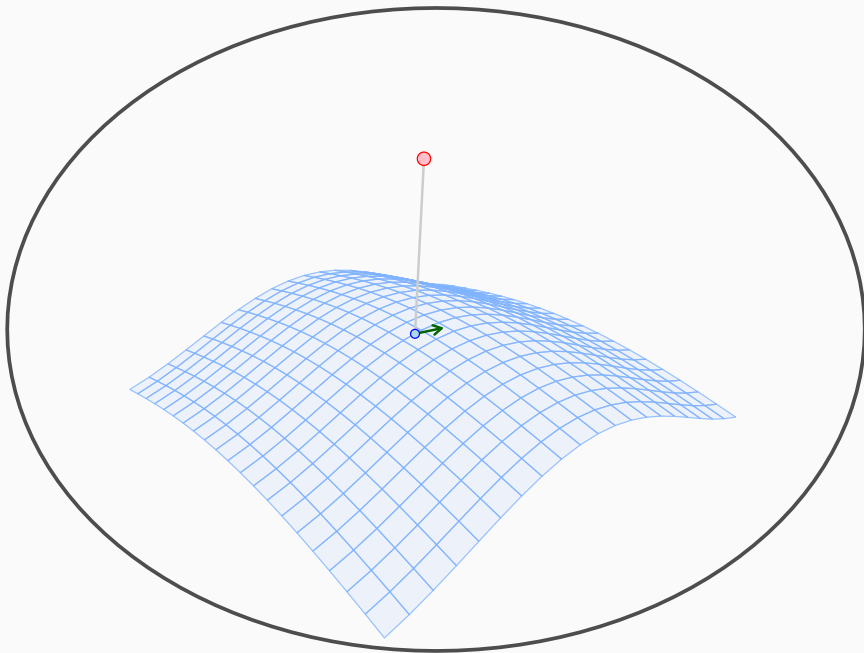


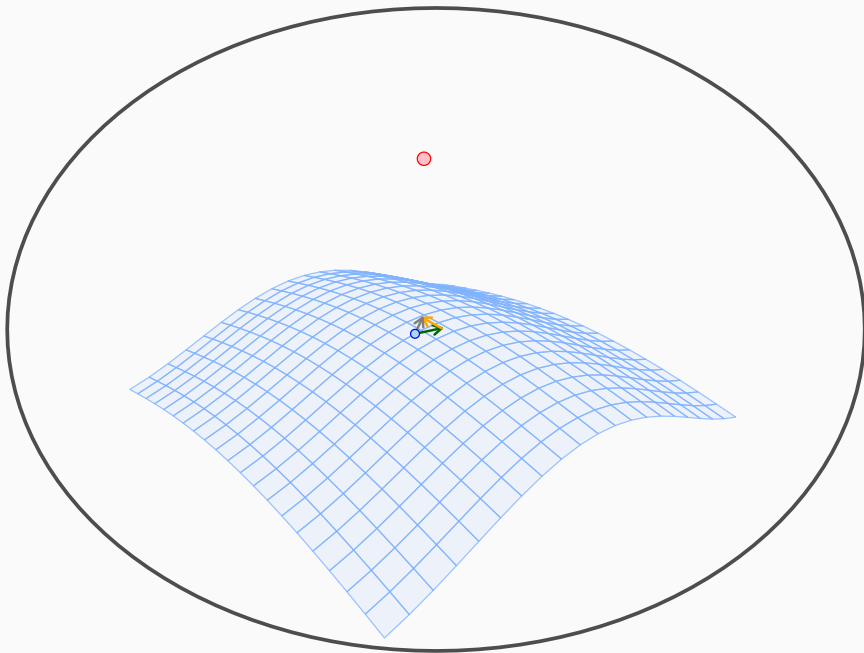


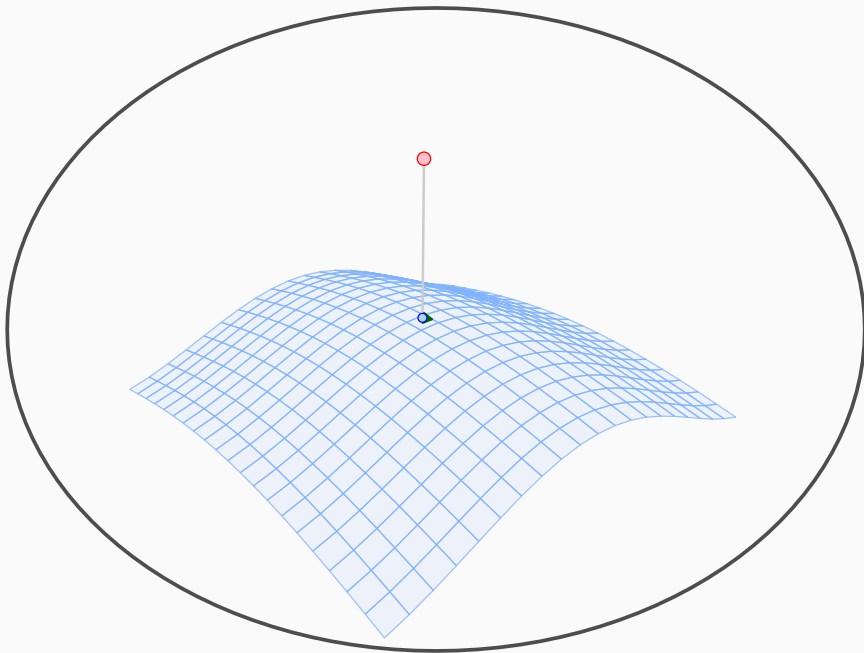


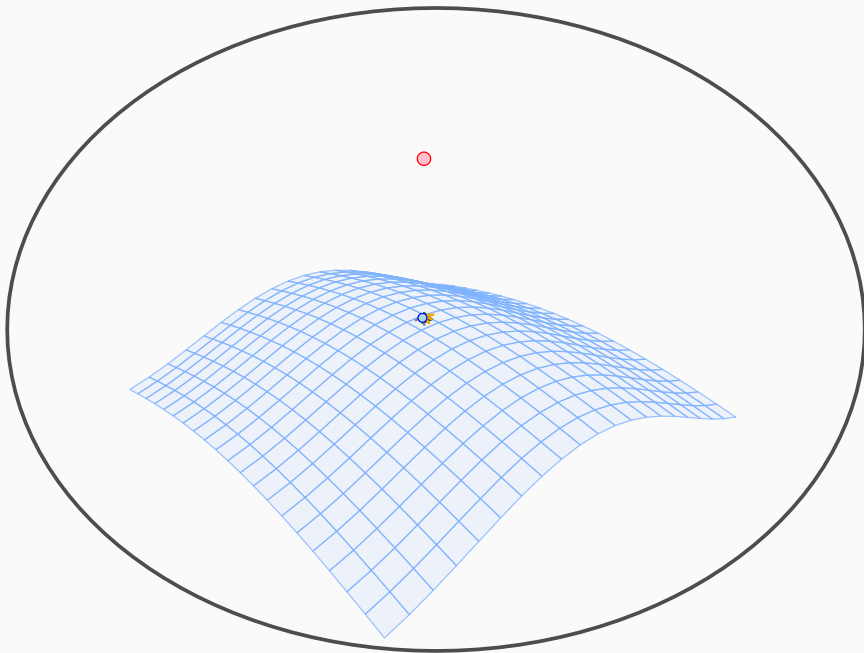




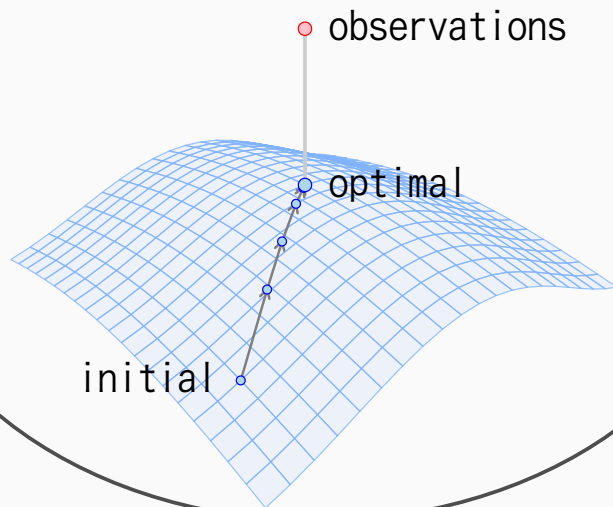








Newton's method



Lemma (Amari, 1967)

$$\begin{aligned}\mathbb{E}^{\theta_{t+1}} [f(\theta_{t+1})] &= \mathbb{E}^{\theta_t} [f(\theta_t)] - \mathbb{E}^{\theta_t} [\nabla f(\theta_t)^\top \Phi_t \nabla L(\theta_t)] \\ &\quad + \frac{1}{2} \text{tr} \mathbb{E}^{\theta_t} [\Phi_t G(\theta_t) \Phi_t^\top \nabla \nabla f(\theta_t)] + \mathcal{O}(\|\Phi_t\|^3)\end{aligned}$$

holds for any smooth function $f(\theta)$, where $\mathbb{E}^\theta$ denotes the expectation with respect to θ , and $G(\theta)$ is defined by

$$G(\theta) = \mathbb{E}_{Z \sim P} [\nabla l(Z; \theta) \nabla l(Z; \theta)^\top] .$$

Definition

Let A be an $m \times m$ square matrix and M be an $m \times m$ symmetric matrix. We define two linear operators as follows:

$$\Xi_A M = AM + (AM)^T,$$

$$\Omega_A M = AMA^T.$$

Lemma

Around the optimal parameter, the following approximated recursive relations for the expectation $\bar{\theta}_t = \mathbb{E}^{\theta_t} [\theta_t]$ and the covariance $V_t = \mathbb{V}^{\theta_t} [\theta_t]$ hold:

$$\bar{\theta}_{t+1} = \bar{\theta}_t - Q_t(\bar{\theta}_t - \theta_*),$$

$$V_{t+1} = V_t - \Xi_{Q_t} V_t + \Omega_{Q_t} V_* - \Omega_{Q_t}(\bar{\theta}_t - \theta_*)(\bar{\theta}_t - \theta_*)^\top,$$

where

$$Q_t = \Phi_t H, \quad V_* = H^{-1} G H^{-1}.$$

(note: $\Xi_A M = A M + (A M)^\top$, $\Omega_A M = A M A^\top$)

Theorem

Let Φ be C/t , where C is a constant matrix. If $\lambda_{\min}(CH) \geq 1$, the leading terms are given by

$$\bar{\theta}_t = \theta_* + S_t(\theta_0 - \theta_*), \quad S_t = \prod_{\tau=2}^t \left(I - \frac{CH}{\tau} \right) = \mathcal{O} \left(\frac{1}{t^{\lambda_{\min}}} \right),$$

$$V_t = \left[(\Xi_{CH} - I)^{-1} \Omega_{CH} \right] \frac{1}{t} V_*, \quad V_* = H^{-1} G H^{-1},$$

where θ_0 is an initial parameter.

Lemma

Let λ_i , $i = 1, \dots, m$ be eigenvalues of A . The eigenvalues of Ξ_A and Ω_A are given by

$$\Xi_A : \lambda_i + \lambda_j, \ i, j = 1, \dots, m,$$

$$\Omega_A : \lambda_i \lambda_j, \quad i, j = 1, \dots, m.$$

Proof.

This follows by the relation

$$\text{cs}(ABC) = (C^T \otimes A) \text{cs}B$$

for any $m \times m$ square matrices A, B, C .



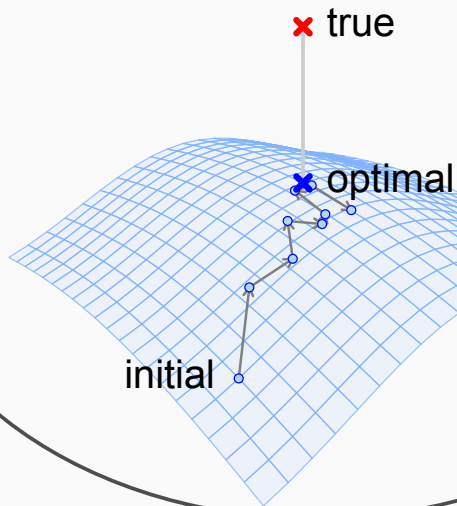
- larger $\lambda_{\min}$ is advantageous to faster convergence of $\bar{\theta}_t$.
- $(\Xi_{CH} - I)^{-1}\Omega_{CH}$ expands V_*/t , which is the minimum covariance attained by batch learning.
- eigenvalues of $(\Xi_{CH} - I)^{-1}\Omega_{CH}$ are given by

$$\frac{\lambda_i \lambda_j}{\lambda_i + \lambda_j - 1},$$

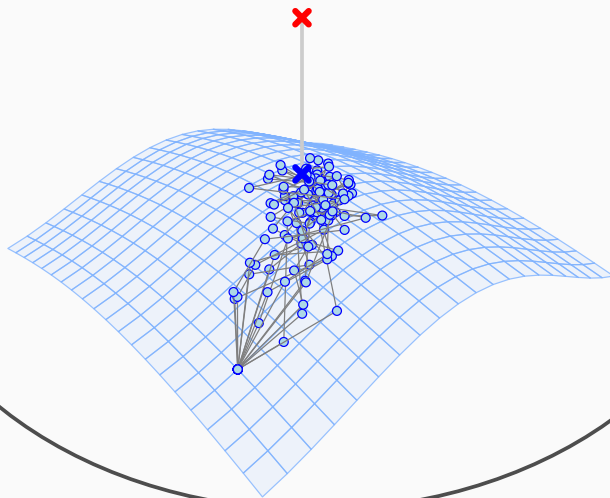
where λ_i 's are eigenvalues of CH .

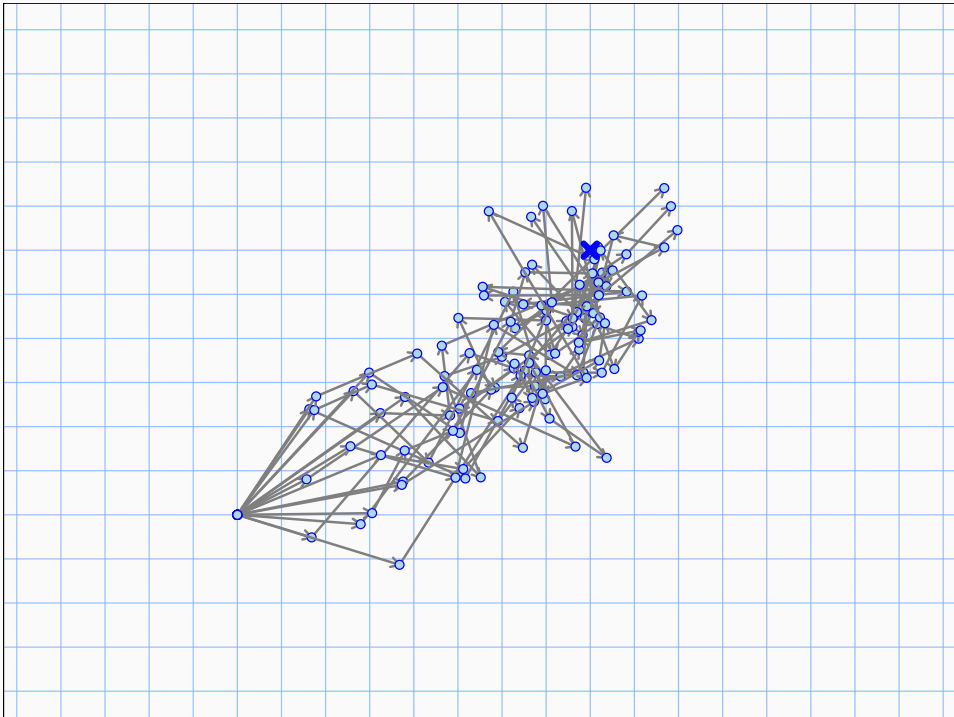
- if $C = H^{-1}$,
all the eigenvalues of $(\Xi_I - I)^{-1}\Omega_I$ are equal to 1, i.e. $V_t = V_*/t$.
- $\Phi_t = H^{-1}/t$ is optimal.

stochastic approximation

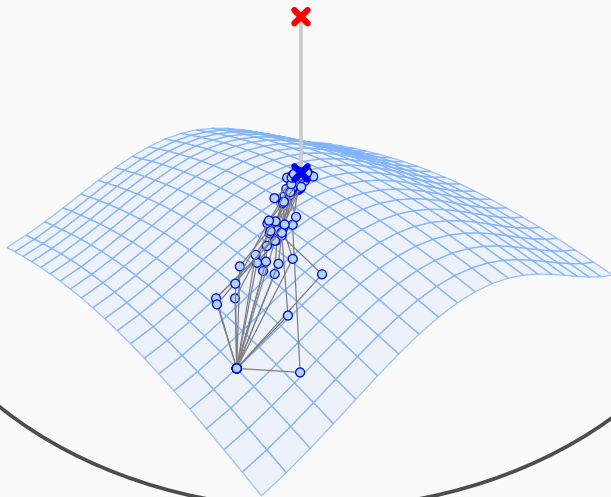


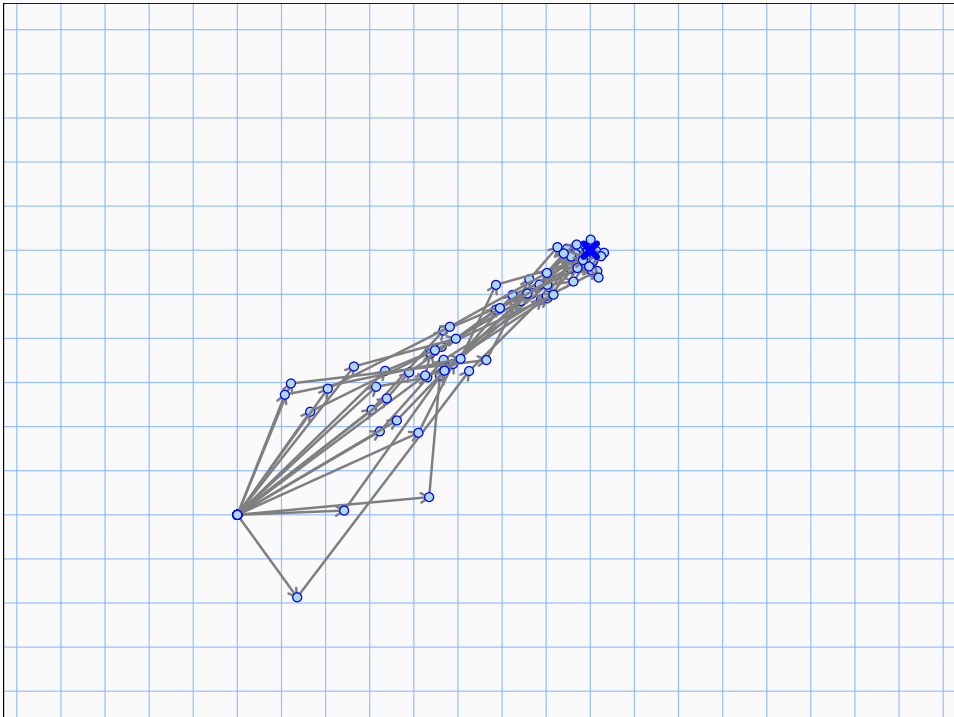
fixed learning rate





optimal learning rate





ILLUSTRATIVE EXAMPLE

a method for evaluating the relative skill levels of players

- Elo rating: Arpad Elo, 1960
used in competitor-versus-competitor games such as chess
scores given to players are updated according to game results
- Glicko rating: Mark Glickman, 1997
including confidence of estimated skill levels
- TrueSkill: Ralf Herbrich et al., 2007
extension to multiplayer games
skill levels are random variables (Bayesian framework)

- score: $\theta = (\theta^1, \theta^2, \dots)$
- event: $z_t = (a \succ b)$ (player a beats player b at time t)
- probability model:

$$\Pr(a \succ b) = P(z_t; \theta) = \frac{1}{1 + \exp(\gamma \cdot (\theta^b - \theta^a))},$$

where γ is defined such that a player whose rating is 200 points greater than the other is expected to have a 75% chance of winning.

- loss function: (negative log loss)

$$l(z_t; \theta) = -\log P(z_t; \theta) = \log(1 + \exp(\gamma \cdot (\theta^b - \theta^a)))$$

fixed learning rate ($k = 32$)



fixed rate: $\Phi_t = \varepsilon I$

- 10 players out of 100
- 20000 games (400[games/pl.])
- $k = 32, 16, 64$
- $\theta_0^i = 1500$

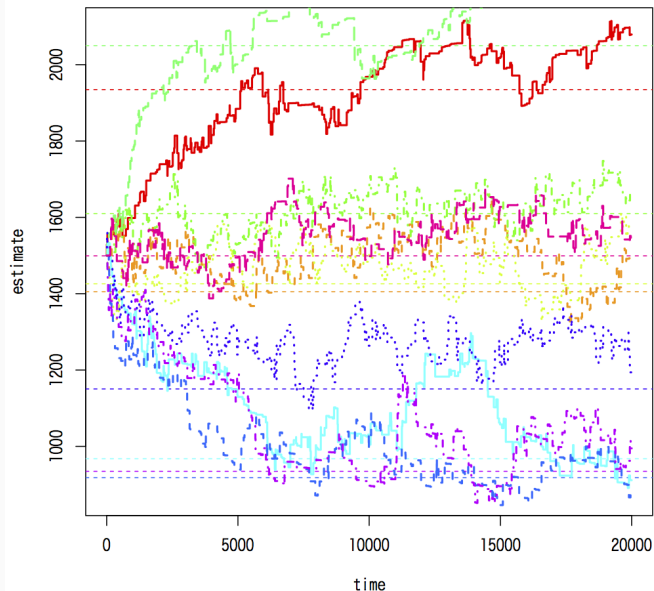
fixed learning rate ($k = 16$)



fixed rate: $\Phi_t = \varepsilon I$

- 10 players out of 100
- 20000 games (400[games/pl.])
- $k = 32, 16, 64$
- $\theta_0^i = 1500$

fixed learning rate ($k = 64$)



fixed rate: $\Phi_t = \varepsilon I$

- 10 players out of 100
- 20000 games (400[games/pl.])
- $k = 32, 16, 64$
- $\theta_0^i = 1500$

- update rule: (Φ : matrix)

$$\theta_t = \theta_{t-1} - \Phi_t \nabla l(z_t; \theta_{t-1}),$$

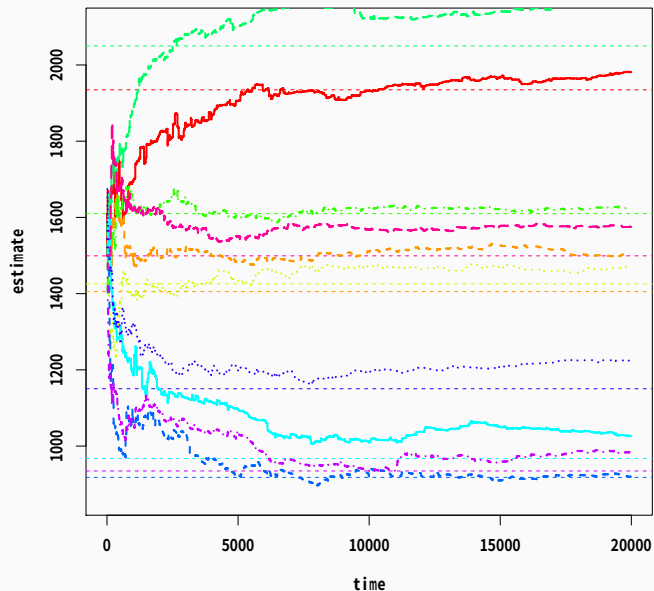
$$\Phi_t = \Phi_{t-1} - \frac{\Phi_{t-1} \nabla l_t \nabla l_t^\top \Phi_{t-1}}{1 + \nabla l_t^\top \Phi_{t-1} \nabla l_t},$$

$$\begin{aligned}\nabla l_t &= \nabla l(z_t; \theta_{t-1}) \\ &= (0, \dots, \underbrace{\gamma(1-P)}_a, \dots, \underbrace{-\gamma(1-P)}_b, \dots, 0)^T\end{aligned}$$

- initial value:

$$\Phi_0 = kl \quad (l \text{ is the identity matrix})$$

optimal learning rate



optimal rate

- 10 players out of 100
- 20000 games (400[games/pl.])
- sensitive to initial value

- original update rule: $\Delta\theta = -\varepsilon\nabla l(z_t; \theta)$
 - only related players are updated: $\Delta\theta^i = 0, i \neq a, b.$
 - sum of θ is kept constant: $\mathbf{1}^\top \Delta\theta = 0.$
- optimal update rule: $\Delta\theta = -\Phi_t \nabla l(z_t; \theta)$
 - all the players are updated, because $\Phi_t = \hat{H}_t^{-1}/t$ is a dense matrix.
 - sum of θ is not necessarily kept constant.
- our problem: design Φ_t to fit the original restriction.

- 1 vs 1 case: (players a and b)

$$\Delta\theta = \alpha \mathbf{a}, \quad \mathbf{a}^\top = \begin{pmatrix} a & b & c \\ 1 & -1 & 0 & \cdots \end{pmatrix},$$

or

$$B^\top \Delta\theta = 0, \quad B^\top = \begin{pmatrix} a & b & c & d \\ 1 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & & & \ddots \end{pmatrix}.$$

- $$\Delta\theta = A\alpha, \quad A^\top = \begin{pmatrix} & a & b & c & d & e \\ 1 & 0 & -1 & 0 & 0 & \cdots \\ 1 & 0 & 0 & -1 & 0 & \cdots \\ 0 & 1 & -1 & 0 & 0 & \cdots \end{pmatrix},$$

$$B^\top \Delta\theta = 0, \quad B^\top = \begin{pmatrix} & a & b & c & d & e & f \\ 1 & 1 & 1 & 1 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 1 & 0 & \cdots \\ 0 & 0 & 0 & 0 & 0 & 1 & \cdots \\ \vdots & \vdots & & & & & \ddots \end{pmatrix}.$$

Problem A

Find an “optimal” gradient $\Delta\theta = \Phi\nabla l(z; \theta)$ subject to

$$\Delta\theta \in \text{Im } A, \quad (\Delta\theta = A\alpha, \alpha \in \mathbb{R}^k)$$

for a matrix $A \in \mathbb{R}^{m \times k}$.

Problem B

Find an “optimal” gradient $\Delta\theta = \Phi\nabla l(z; \theta)$ subject to

$$\Delta\theta \in \text{Ker } B^\top, \quad (B^\top \Delta\theta = 0)$$

for a matrix $B \in \mathbb{R}^{m \times (m-k)}$,

cf. $f(\theta) = \text{const.} \Rightarrow \nabla f(\theta)^\top \Delta\theta = 0$

- optimality is defined in terms of

$$\text{minimize } \|H^{-1}\nabla l - \Delta\theta\|_M,$$

where $\|x\|_M^2 = \langle x, x \rangle_M$ and $\langle x, y \rangle_M = \langle Mx, y \rangle$.

- M is chosen as H , because
 - quadratic approximation of population loss:

$$\|\theta - \theta_*\|_H^2 = (\theta - \theta_*)^\top H(\theta - \theta_*) = L(\theta) - L(\theta_*)$$

- Mahalanobis distance in maximum likelihood case:

$$\mathbb{V}[\hat{\theta}_t] = \frac{1}{t}H^{-1}GH^{-1} = \frac{1}{t}H^{-1}$$

- decompose Φ_t into scalar and matrix parts as

$$\Phi_t = \varepsilon_t C, \quad (\text{e.g., } \varepsilon_t = 1/t)$$

- solutions for the problems are:

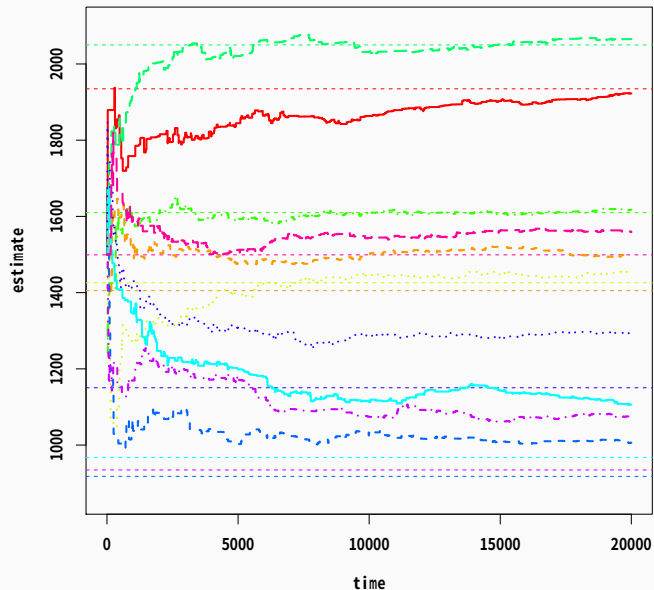
Problem A

$$C_A = A(A^T H A)^{-1} A^T$$

Problem B

$$C_B = H^{-1} - H^{-1} B (B^T H^{-1} B)^{-1} B^T H^{-1}$$

sub-optimal learning rate



sub-optimal rate

- 10 players out of 100
- 20000 games (400[games/pl.])

- C_A and C_B are symmetric (only when $M = H$).
- $C_A H$ or $C_B H$ is a projection matrix:

$$\lambda = \begin{cases} 1, & v \in \text{Im } A \text{ or } \text{Ker } B, \\ 0, & \text{otherwise.} \end{cases}$$

- if k is small, calculating C_A is more efficient than C_B .
- only a few parameters are updated, however convergence is as good as optimal case.
(information loss is quite small in some case)

CONCLUSION

we have investigated

- dynamics of convergence phase of on-line learning,
- conditions for optimal convergence rate,
- optimal projection of gradients to subspaces,

practical applications would be

- skill level rating systems,
- on-line learning for Bradley-Terry model,
- distributed control systems.

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 Murata, Noboru and Shun-ichi Amari (Apr. 1999). “Statistical analysis of learning dynamics.” In: *Signal Processing* 74 (1), pp. 3–28. DOI: [10.1016/S0165-1684\(98\)00206-0](https://doi.org/10.1016/S0165-1684(98)00206-0).